

Pradyut Kumar

Purdue University, West Lafayette, IN
Ph.D. Candidate, Ecology and Evolutionary Biology

pradyutkumar01@gmail.com
linkedin.com/in/pradyut-kumar
pradyutk42.github.io

EDUCATION	Purdue University , West Lafayette, IN2024–Present Ph.D. in Biological Sciences, Ecology and Evolutionary Biology Advisor: Prof. Qixin He Birla Institute of Technology and Science, Pilani , Pilani, India2019–2024 M.Sc. Biological Sciences and B.E. Electronics and Instrumentation Advisor: Prof. Vishweshha Guttal, Indian Institute of Sciences
RESEARCH INTERESTS	My research interests lie at the intersection of infectious disease ecology, evolution, and epidemiology, with a particular focus on the emergence and spread of antimalarial drug resistance. I use agent-based modeling, population genetics, and mechanistic approaches to investigate how ecological and evolutionary processes shape pathogen dynamics. I am also interested in developing predictive models that connect biological mechanisms with population-level patterns of disease and resistance.
RESEARCH EXPERIENCE	Antimalarial Drug Resistance Evolution Jan. 2025–Present <i>Purdue University, West Lafayette, IN</i> Advisor: Prof. Qixin He Develop agent-based models of <i>Plasmodium falciparum</i> transmission to study how different ecological factors and selection pressures shape anti-malarial drug resistance establishment and persistence. Quantify invasion probability and resistance dynamics across single- and multi-locus architectures and identify predictors of outcomes using surrogate Gaussian Process models. Quantitative Study of Animal Behaviour June 2023–Dec. 2023 <i>Indian Institute of Science, Bengaluru, India</i> Advisor: Prof. Vishweshha Guttal Used image processing and quantitative methods to study collective behavior in mixed-species fish schools. Evaluated multi-animal tracking pipelines and developed an optical-flow approach for large fish schools. Mycoremediation of Dye-Containing Wastewater Sept. 2022–May 2023 <i>BITS Pilani, Pilani, India</i> Advisor: Prof. Jitendra Panwar Screened fungi for dye decolorization and measured degradation using UV-Vis, GC/MS, and SDS-PAGE. Analysis of Land-Use Patterns in the Gangetic Plain June 2022–Aug. 2022 <i>NITI Aayog</i> Advisor: Dr. Neelam Patel Analyzed decadal district-level land-use patterns and strategies relevant to agricultural productivity.
CONFERENCE POSTERS	Immune Selection & Co-Infection Decouple Antimalarial Resistance Establishment & Long-Term Persistence Ecology and Evolution of Infectious Diseases Conference (EEID)June 2026
TEACHING EXPERIENCE	Purdue University , West Lafayette, INFall 2025 <i>Teaching Assistant, BIOL 41919: Data Science for Biologists</i> Supported introductory R instruction; graded and designed assignments and assessment questions. <i>Teaching Assistant, BIOL 13500: Introduction to Biology Laboratory</i> 2024–2026 Guided laboratory and field research, including a study of soil microbiomes across ecological conditions.
MENTORING	Undergraduate Research Mentor , Purdue UniversitySummer 2025–Present Mentor a Universidad de los Andes undergraduate on antimalarial resistance modeling through Purdue’s SURF program and beyond.
AWARDS & HONORS	Purdue Graduate Student Government (PGSG) Travel Grant, \$5002026 Rosenberg Award, Purdue University, \$4,2502024
PROFESSIONAL EXPERIENCE	Software Engineering Intern Jan. 2024–June 2024 <i>Aurigo Software Technologies, Bengaluru, India</i> Developed reports and dynamic forms for Aurigo Masterworks using C#, XML, JavaScript, and .NET.
TECHNICAL SKILLS	Programming: Julia, R, Python, C/C++, MATLAB Scientific Computing: Agent-based and statistical modeling, simulation, data analysis, HPC Tools: Git, L ^A T _E X, SLURM, Excel Imaging / Visualization: idtracker.ai, TRex, Blender